ECE 4704 Advances in Computer Vision

Course Project Ideas

Teams of at most three students. Each project is scoped at about 20 working hours per student, which is roughly 60 team-hours in total. Every project below is a working application, and every one of them runs on Google Colab.

Rules, milestones and deliverables

Teams are formed and an idea is chosen in Week 2. Three graded milestones follow. In Week 7 (3 marks) the team submits a proposal, the prepared dataset and a first baseline. In Week 12 (2 marks) the team submits a progress report with the main model results. In Week 15 (10 marks) the team submits the final work and presents a live demo.

Each project description below has a course path with three stages. The first stage is a logistic regression or MLP baseline, which is the Week 7 milestone and uses only the lectures of Weeks 2 to 6. The second stage is the main model, which is the Week 12 milestone. The third stage is the application itself, which is what gets demonstrated in Week 15.

The baseline is not busy work. It is what makes the final number mean something, because a CNN at 94 percent is only impressive next to a baseline at 71 percent on the same data.

A workable split of the 60 team-hours is about 20 hours for data preparation and setup, 25 hours for experiments and 15 hours for the report and the demo. Any project that needs more than a few hours of GPU time is too large for this course.

Teams should not reuse the lab dataset for the same purpose, since the labs already cover face recognition and facial expression end to end.

A. Recognition applications

These can start immediately after Week 2, because the baseline stage needs nothing beyond logistic regression.

1. Drowsy driver alert

A camera watches the driver and sounds an alarm when the eyes stay closed or yawning becomes frequent. This is a real safety product and it demonstrates well in front of a class.

Course path. Logistic regression and an MLP on cropped eye images for open against closed, then a CNN, then transfer learning, then a loop over webcam frames with a time window that decides when to raise the alarm.

Data and demo. MRL Eye dataset, YawDD, or images the team records themselves. The demo is a laptop webcam with the alarm sounding live.

2. Sign language alphabet translator

Point a camera at a hand and the system prints the letter, building words in real time. Choose Arabic sign language if a dataset is available, otherwise American.

Course path. MLP on flattened images for a few letters, then a CNN on all of them, then transfer learning, then a real-time loop with letter smoothing so a held pose types one letter and not twenty.

Data and demo. ASL Alphabet or ArASL datasets, plus a set the team records to test on new hands and new lighting. The demo is spelling a word live.

3. Attendance without a register

A camera at the door recognises who walked in and writes the attendance sheet. The engineering interest is that new students must be enrolled from two or three photos, without retraining the network.

Course path. Logistic regression on extracted features for a closed set of people, then a CNN classifier, then an embedding model trained with pairs or triplets so that enrolment becomes storing one vector per person.

Data and demo. LFW plus photos of the team and their classmates. Report an ROC curve and the chosen threshold, and show the false accept and false reject cases honestly.

4. Crop disease diagnosis from a phone photo

A farmer photographs a leaf and the app names the disease and its severity. This is one of the few projects here with real social value in Egypt.

Course path. MLP on colour and texture features, then a CNN, then transfer learning, then a test on photographs taken outdoors rather than in the laboratory conditions of the training set.

Data and demo. PlantVillage for training, plus real field photographs the team takes. The interesting result is usually the accuracy drop from clean data to real photographs.

5. Recycling sorter

A camera above a bin decides whether an item is plastic, glass, metal, paper or organic waste, which is the vision half of an automatic sorting machine.

Course path. Logistic regression on colour histograms, then a CNN, then transfer learning, then a speed and size study so the model could sit on a small device next to a conveyor.

Data and demo. TrashNet or TACO, plus items the team photographs on their own desk. The demo is holding objects in front of the camera one at a time.

6. Traffic sign recognition for driver assistance

The system reads speed limits and warning signs from a dashboard camera and displays the current limit, which is the feature already shipping in modern cars.

Course path. MLP on small sign crops, then a CNN, then transfer learning, then a robustness study under blur, rain, night and partial occlusion, since that is where the real failures happen.

Data and demo. GTSRB for training and video recorded from a car for testing. The demo is a processed dashcam clip.

7. A reading assistant for the visually impaired

The phone camera identifies banknotes, medicine boxes or product packages and speaks the answer aloud, which is genuinely useful and easy to show.

Course path. Logistic regression on features for a few classes, then a CNN, then transfer learning, then the audio output and a confidence threshold so the app stays silent rather than guessing wrongly.

Data and demo. Photographs of Egyptian banknotes or common packages collected by the team, deliberately including folded notes and bad lighting.

8. A screening assistant for medical images

Skin lesion triage or chest X-ray screening, presented as a tool that flags cases for a doctor rather than one that diagnoses.

Course path. Logistic regression on features, then a CNN, then transfer learning, then Grad-CAM heat maps so every prediction comes with a visible reason, plus a report on sensitivity at a chosen operating point.

Data and demo. HAM10000 for skin lesions or a public chest X-ray set. The report must discuss class imbalance and the cost of a missed positive.

B. Restoration and generation applications

These use the autoencoder, VAE and GAN lectures. Teams should still build the baseline stage early.

9. Old family photo restoration

Feed in a scratched, noisy, faded photograph and get back a clean one, then colourise it. Every team member will run their own grandparents photographs through it, which is what makes this project popular.

Course path. An MLP autoencoder on small patches, then a deep convolutional denoising autoencoder trained on artificially damaged images, then colourisation. Report PSNR and SSIM against the clean originals.

Data and demo. Any face or scene dataset with damage added synthetically for training, and genuinely old photographs for the demo, shown side by side.

10. Factory defect inspector

A camera on a production line flags defective items. The twist is that defects are rare, so the model is trained only on good items and flags anything it cannot reconstruct.

Course path. An MLP classifier on the few labelled defects as a baseline, then an autoencoder trained only on good samples, using reconstruction error as the anomaly score, then a threshold chosen from a precision and recall curve.

Data and demo. MVTec AD, or photographs of a local product such as dates, tiles or fabric. The demo shows the error heat map pointing at the defect.

11. Selfie to cartoon

Turn a photograph of a face into a cartoon or an oil painting, which is the filter every phone app now has.

Course path. An autoencoder that reconstructs faces as a baseline, then CycleGAN between the photograph domain and the cartoon domain, then a study of which failures appear, such as glasses, beards and unusual poses.

Data and demo. CelebA with a cartoon set such as Cartoon Set or anime faces. The demo is running the team photographs through it live.

12. Catching the fakes

Build a generator of fake faces, then build the detector that catches them, and see which one wins. This is the most current topic in the course and it needs no external dataset of fakes.

Course path. An MLP and then a CNN trained to separate real from generated faces, using images the team generates with their own DCGAN, then a test on fakes from a different generator to see whether the detector transfers.

Data and demo. CelebA for the real images and the team GAN for the fakes. The demo is a page where the audience guesses and then the detector answers.

13. Enrolling a new face from three photographs

The attendance system from project 3 fails when a new student gives only three photographs, so this project generates more of them and measures whether it actually helps.

Course path. An MLP classifier on the three photographs as a baseline, then a conditional GAN or a VAE that generates variations in pose and lighting, then a measurement against ordinary augmentation such as flips and crops.

Data and demo. Any face dataset, artificially reduced to three images per person. A negative result that is properly measured is a valid outcome.

C. Segmentation applications

14. Virtual background for video calls

Separate the person from the background in a live video and replace the background, which is the feature everyone used during the pandemic and almost nobody has implemented.

Course path. A per-pixel MLP on colour as a baseline, which will fail in an instructive way, then an FCN, then U-Net, then a frame rate study to see how close to real time it can run.

Data and demo. A human segmentation dataset for training, and the team webcam for the demo, with background replacement and blur both shown.

15. Measuring a tumour instead of describing it

The system outlines the tumour in every MRI slice and reports its area and volume, so that growth between two scans becomes a number rather than an opinion.

Course path. A per-pixel baseline, then U-Net, then a comparison of cross-entropy against Dice loss and the effect of augmentation, then the volume computation across slices.

Data and demo. A public brain MRI segmentation set. Report mean IoU and Dice, and show the worst failures rather than only the best cases.

16. Rooftop solar survey from satellite images

Segment building rooftops across a district and estimate the usable area for solar panels, producing a map that a planner could actually use.

Course path. A pixel classifier baseline, then U-Net for rooftop segmentation, then conversion of pixels to square metres, then a district level estimate with its assumptions stated.

Data and demo. Inria Aerial Image Labeling or an open buildings dataset. The demo is a coloured map of one neighbourhood.

17. Road and lane segmentation for driver assistance

Mark the drivable road and the lane markings in a dashcam video, which is the first stage of a lane departure warning.

Course path. A colour based baseline, then FCN, then U-Net, then a test on night and rain footage where the baseline collapses and the network degrades more gracefully.

Data and demo. KITTI Road or Cityscapes for training, plus video recorded locally. The demo is a processed clip with a lane departure alert.

D. Detection and video applications

18. Parking gate that reads plates

The camera at the gate finds the licence plate, reads it, and opens the barrier for a registered vehicle. This is the same design as the parking lot question in the exams.

Course path. A classifier on cropped plates as a baseline, then a detector with a classification head and a bounding box regression head, then a fine-tuned YOLO, then character recognition and the gate decision.

Data and demo. An open plate dataset plus photographs of Egyptian plates taken by the team. The demo is a video of cars arriving at a gate.

19. Site safety monitor

A camera on a construction site or in a laboratory flags anyone without a helmet, vest or goggles, and counts the violations per hour.

Course path. A classifier on cropped people as a baseline, then a fine-tuned detector for person and equipment, then the rule that links them, then a violation log with timestamps.

Data and demo. A hard hat or PPE detection dataset, tested on footage the team records. The demo is a processed video with the log beside it.

20. Counting traffic at an intersection

Count the cars, buses and motorbikes passing in each direction and produce the hourly profile a traffic engineer would ask for.

Course path. A vehicle classifier on crops as a baseline, then a fine-tuned detector, then simple tracking across frames so that one car is counted once, then the traffic report.

Data and demo. Video recorded from a window or a public traffic camera. The demo is the counting overlay plus the resulting graph.

21. Shelf audit for a supermarket

A photograph of a shelf reports which products are present and where the gaps are, which is a service that real companies sell.

Course path. A product classifier on crops as a baseline, then a detector fine-tuned on shelf images, then gap detection and a stock report per shelf.

Data and demo. SKU-110K or the Grocery datasets, plus photographs from a local shop with permission. The demo is an annotated shelf photograph.

22. How full is the lab right now

Count the people in a room or a queue from a ceiling camera and publish an occupancy number, which is the campus version of a crowd monitoring system.

Course path. A classifier on image patches as a baseline, then a fine-tuned person detector, then a comparison with a density map approach when the crowd gets dense enough that detection starts to fail.

Data and demo. A crowd counting dataset for training, and footage of a real campus space for the demo, with the counting error reported honestly.